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Advanced Patterns & Complete Workflows

***SESSION 2**: From Simple Templates → Orchestrated Systems*

ReAct + Tree of Thoughts + Live Java Demo

***Duration**: 60 minutes | **Format**: 15min lecture + 40min hands-on + 5min wrap*

Today's Agenda

Time	Activity	Type
0-5 min	Session 1 Recap + Advanced Patterns	📊 **LECTURE**
5-10 min	Spec-Kit Methodology	📊 **LECTURE**
10-15 min	ReAct + Tree of Thoughts Overview	📊 **LECTURE**
15-25 min	🎯 **DEMO: Interview Prep Workflow**	🔴 **HANDS-ON**
25-40 min	🎯 **YOUR TURN: Build Interview Materials**	🔴 **HANDS-ON**
40-55 min	🎯 **LIVE JAVA DEMO: Spring Migration**	🔴 **HANDS-ON**
55-60 min	Integration + Next Steps	📊 **LECTURE**

Session 1 Recap: Foundation Established

You mastered:

✅ **Three Approaches Framework** (ADRs vs Structured vs Tool-assisted)
✅ **Tier Evaluation** (proven vs emerging vs experimental)
✅ **Foundational Patterns** (Persona, Few-shot, Template, Chain-of-Thought)
✅ **Real application** (Priority Builder with actual deliverables)

Today's evolution:

Simple patterns → **Orchestrated workflows**
Single-step tasks → **Multi-phase reasoning**
Generic applications → **Cross-domain scaling**

Advanced Patterns (Research-Backed)

When simple patterns aren't enough for complex tasks:

🔄 **ReAct Pattern** (Yao et al., 2022)

Format: Think → Act → Observe → Think → Act...

Use: Multi-step tasks with validation checkpoints

🌳 **Tree of Thoughts** (Yao et al., 2023)

Format: Generate alternatives → Evaluate → Choose best

Use: Decision points with tradeoffs

📁 **Spec-Kit Methodology** (Your proven approach)

Format: 3-4 structured files working together

Use: Complex tasks requiring systematic breakdown

Spec-Kit 4-File Workflow

Spec-Kit 4-File Workflow

Key advantage: Separation of concerns enables reuse and systematic quality

Pattern Integration: Interview Strategy Example

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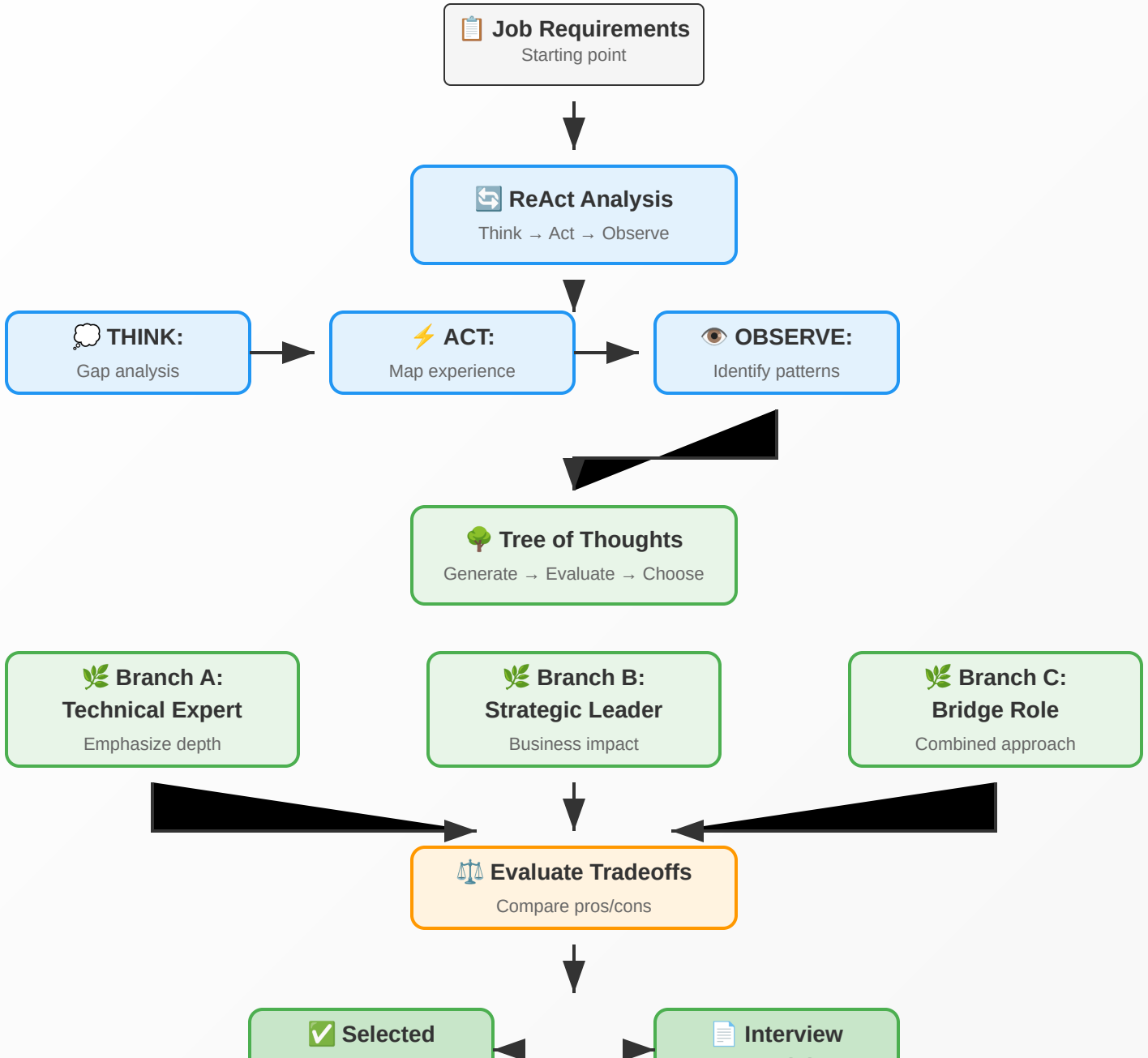

🎯 HANDS-ON TIME

Next 40 minutes: Advanced patterns in practice

Multi-step reasoning + live technical demo

🎯 DEMO: Interview Prep Workflow (10 minutes)

Scenario: "Senior Manager - AI Strategy" role preparation

Watch the 4-file workflow:

knowledge-base.md (2 min): Interview fundamentals, STAR method, positioning strategies

specification.md (3 min): This role + candidate background analysis

implementation-plan.md (3 min): ReAct analysis + Tree of Thoughts evaluation

execution.md (2 min): Generated positioning + one-pager + talking points

Key insight: Each file has specific purpose, builds on previous files

🎯 YOUR TURN: Build Interview Strategy (15 minutes)

Materials: Choose from provided job descriptions in `job-descriptions.md` + your demo persona

Phase 1 (8 min): Create specification.md

Use provided knowledge-base.md (interview fundamentals)

Select job description from: Senior Manager AI Strategy, Principal Consultant Banking, Director Digital Transformation, or Senior Principal Cloud Architecture

Document target role and your background using chosen demo persona

Identify gaps and strengths for positioning

Phase 2 (7 min): Execute implementation-plan.md

ReAct analysis: Think → Act → Observe your positioning options

Tree of Thoughts: Generate 3 strategies → Evaluate → Choose best

Document your reasoning and selected approach

Success criteria: Clear positioning strategy with rationale

🖥️ LIVE TECHNICAL DEMO

Same patterns → Code generation

Systematic Spring Boot migration

🎯 LIVE JAVA DEMO: Spring Migration (15 minutes)

I'll screen share my system and demonstrate:

Setup (2 min): Spring Boot 2→3 migration challenge

javax → jakarta namespace changes

Annotation modernization

Security configuration updates

Three Approaches Live (13 min):

ADRs + Config (3 min): `.github/copilot-instructions.md` approach

Structured Files (5 min): spec/ folder workflow execution

Tool-Assisted (5 min): Windsurf cascade workflow (290 lines)

Key insight: Same systematic thinking scales from priorities → interviews → code

What You're Seeing: Patterns in Code

ReAct Pattern in Action:

Think: "Dependencies must be updated before annotations"

Act: Update javax → jakarta imports first

Observe: `mvn compile` - success, proceed to next phase

Tree of Thoughts for Decisions:

Security Config Option A: Keep current (low risk)

Security Config Option B: Modernize to SecurityFilterChain

Security Config Option C: Full OAuth2 rewrite

Decision: Option B (balanced modernization)

Spec-Kit Structure: knowledge-base → specification → implementation-plan

Cross-Domain Pattern Recognition

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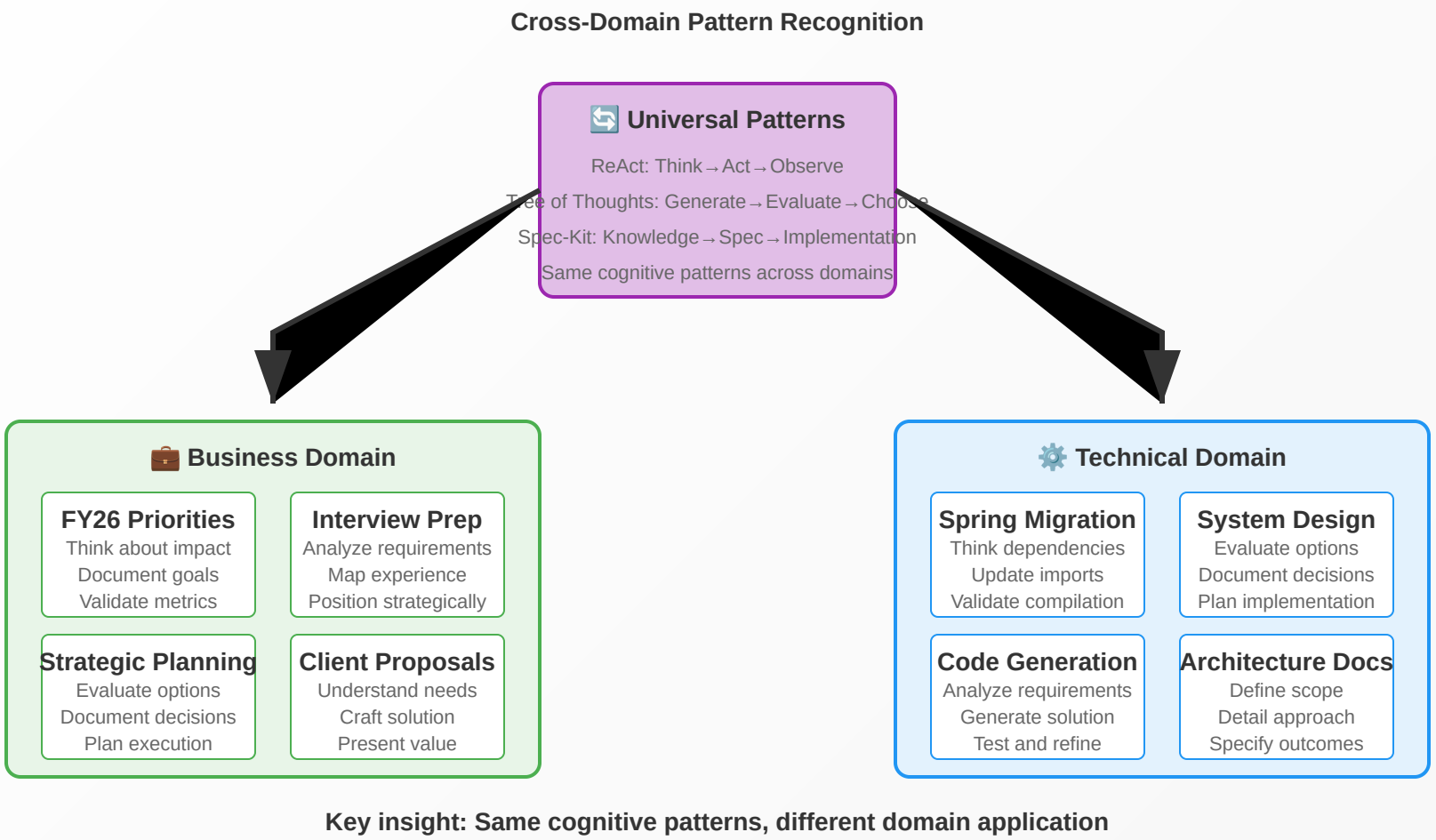

Key insight: Same cognitive patterns, different domain application

Key insight: **Same cognitive patterns**, different domain application

When to Use Advanced vs Simple Patterns

Simple Patterns (Session 1):

✅ Single-step tasks

✅ Well-understood domains

✅ Template-driven outputs

✅ Quick iterations needed

Advanced Patterns (Session 2):

✅ Multi-step reasoning required

✅ Decision points with tradeoffs

✅ Complex domain knowledge

✅ Audit trail needed

✅ Team scalability important

Decision rule: Use simplest approach that handles the complexity

Complete Toolkit: What You Now Have

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Evaluation Framework:

Three Approaches (ADRs vs Structured vs Tool-assisted)

Tier Assessment (proven vs emerging vs experimental)

Pattern Library:

Foundational: Persona, Few-shot, Template, Chain-of-Thought

Advanced: ReAct, Tree of Thoughts, Spec-Kit workflows

Real Applications:

Priority Builder methodology

Interview preparation system

Technical workflow orchestration

Cross-domain scaling: Same patterns work for business + technical tasks

Taking This Forward

This week:

Apply ReAct pattern to one complex work decision

Use Tree of Thoughts for your next strategic choice

Create spec-kit workflow for a repeated task

Longer term:

Evaluate new tools using Tier Framework

Build team templates using proven patterns

Scale systematic thinking across your domain

Remember: Patterns > formats. Choose based on complexity and team needs.

Resources & Materials

Keep forever:

Three Approaches decision matrix

Foundational + Advanced pattern reference

Priority Builder Agent (325 lines)

Interview prep 4-file templates

Spring migration workflow examples

Continue learning:

White et al. (2023): Prompt Pattern Catalog

Yao et al. (2022): ReAct paper

Yao et al. (2023): Tree of Thoughts paper

You're Now Systematic Prompt Engineers

From ad-hoc prompting → methodical workflows

Apply these patterns to create exceptional work

Questions? Discussion? Feedback?